Offline POMDP Algorithms

Break

So far, we have only dealt with n-step value functions, if we want to deal with infinite-horizon discounted POMDPs, how can we construct an infinite-horizon α vector to start with?

Always take action a $\vec{R}^a$

$$\alpha^a = \vec{U}^a = (I - \gamma T^a)\vec{R}^a$$

Infinite-Horizon POMDP Lower Bound Improvement (Value Iteration)

set of α vectors

evaluating always taking a $\forall a \in A$

$\Gamma \leftarrow$ blind lower bound

loop

$\Gamma \leftarrow \Gamma \cup \mathrm{backup}(\Gamma)$

$\Gamma \leftarrow \mathrm{prune}(\Gamma)$

Complexity $O\left(|\Gamma|\,|A|\,|O|\,|S|^2 + |A|\,|S|\,|\Gamma|^{|O|}\right)$

Shani G, Pineau J, Kaplow R. A survey of point-based POMDP solvers. *Auton Agent Multi-Agent Syst*. 2013;27(1):1-51. doi:10.1007/s10458-012-9200-2

Infinite-Horizon POMDP Lower Bound Improvement (Value Iteration)

$\Gamma \leftarrow$ blind lower bound

loop

$\quad \Gamma \leftarrow \Gamma \cup \mathrm{backup}(\Gamma)$

$\quad \Gamma \leftarrow \mathrm{prune}(\Gamma)$

$\mathrm{backup}(\Gamma)$

$$\alpha^{a,o}[s] = \sum_{s'} Z(o \mid a, s')\, T(s' \mid s, a)\, \alpha[s'] \quad \forall\, a, o, s, \alpha \in \Gamma$$

$$\Gamma^{a,o} = \left\{ \frac{1}{|O|}\, r_a + \alpha^{a,o} \;:\; \alpha \in \Gamma \right\} \quad \forall\, a, o$$

$$\Gamma^{a} = \bigoplus_{o \in O} \Gamma^{a,o} \quad \forall\, a$$

$$\Gamma' = \bigcup_{a \in A} \Gamma^{a}$$

Complexity $O\left(|\Gamma|\,|A|\,|O|\,|S|^2 + |A|\,|S|\,|\Gamma|^{|O|}\right)$

Shani G, Pineau J, Kaplow R. A survey of point-based POMDP solvers. *Auton Agent Multi-Agent Syst*. 2013;27(1):1-51. doi:10.1007/s10458-012-9200-2

Point-Based Value Iteration (PBVI)

Maintain belief set B, α-vector set Γ

repeatedly call point_backup for each $b \in B$

each point backup adds 1 new α vector

point_backup(Γ, b)

for $a \in A$

for $o \in O$

belief update

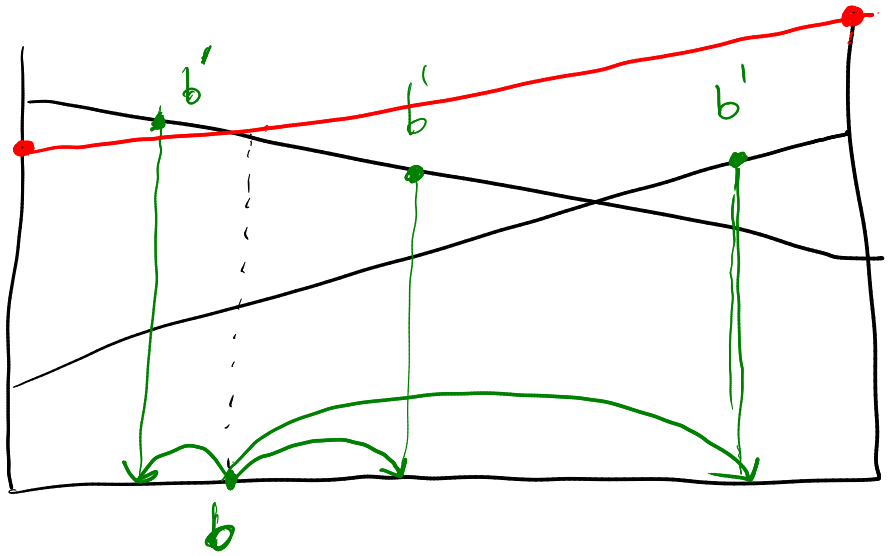
$b' \leftarrow \tau(b, a, o)$

$\alpha_{a,o} \leftarrow \underset{\alpha \in \Gamma}{\operatorname{argmax}}\ \alpha^\top b'$

for $s \in S$

$\boxed{\alpha_a[s]} = R(s,a) + \gamma \sum_{s',o} T(s' \mid s, a)\, Z(o \mid a, s')\, \alpha_{a,o}[s']$

return $\underset{\alpha_a}{\operatorname{argmax}}\ \alpha_a^\top b$



$$O\left(|\Gamma|\,|A|\,|O|\,|S|^2 + |A|\,|S|\,|\Gamma|\,|B|\right)$$

Shani G, Pineau J, Kaplow R. A survey of point-based POMDP solvers. *Auton Agent Multi-Agent Syst*. 2013;27(1):1-51. doi:10.1007/s10458-012-9200-2

SARSOP

Successive Approximation of Reachable Space under Optimal Policies

(One instantiation of PBVI)

- Can solve problems with up to ~100,000 states
- Hidden implementation details :(((
- HSVI - similar algorithm, better paper

Kurniawati H, Hsu D, Lee WS. SARSOP: Efficient Point-Based POMDP Planning by Approximating Optimally Reachable Belief Spaces.

SARSOP

Successive Approximation of Reachable Space under Optimal Policies

(One instantiation of PBVI)

- Start with $B = \{b_0\}$, Γ = blind lower bound

- Can solve problems with up to ~100,000 states
- Hidden implementation details :(((
- HSVI - similar algorithm, better paper

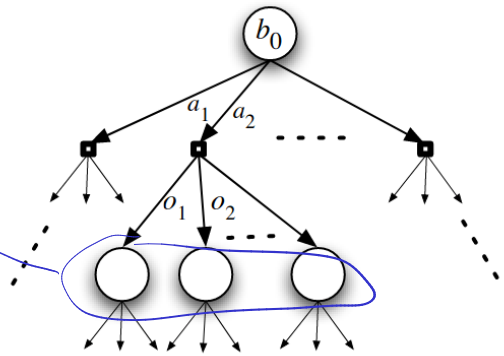
Kurniawati H, Hsu D, Lee WS. SARSOP: Efficient Point-Based POMDP Planning by Approximating Optimally Reachable Belief Spaces.

SARSOP

Successive Approximation of Reachable Space under Optimal Policies

(One instantiation of PBVI)

- Start with $B = \{b_0\}$, Γ = blind lower bound
- Maintain B with a belief tree



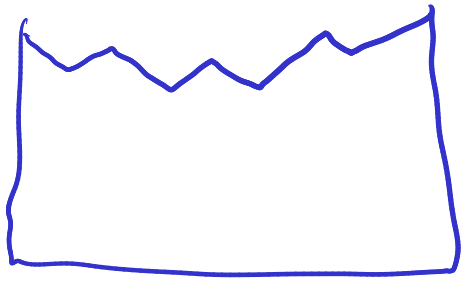
- Can solve problems with up to ~100,000 states
- Hidden implementation details :(((
- HSVI - similar algorithm, better paper

Kurniawati H, Hsu D, Lee WS. SARSOP: Efficient Point-Based POMDP Planning by Approximating Optimally Reachable Belief Spaces.

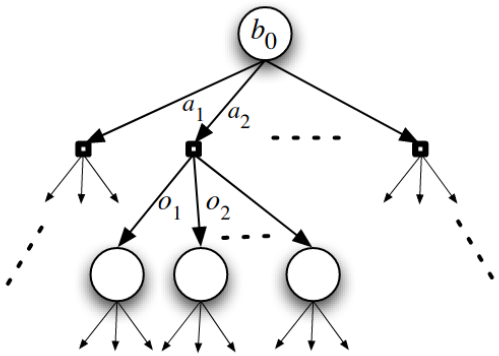
SARSOP

Successive Approximation of Reachable Space under Optimal Policies

(One instantiation of PBVI)



- Start with $B = \{b_0\}$, Γ = blind lower bound
- Maintain B with a belief tree
- Maintain value bounds (Γ: lower, "sawtooth": upper)



- Can solve problems with up to ~100,000 states
- Hidden implementation details :(((
- HSVI - similar algorithm, better paper

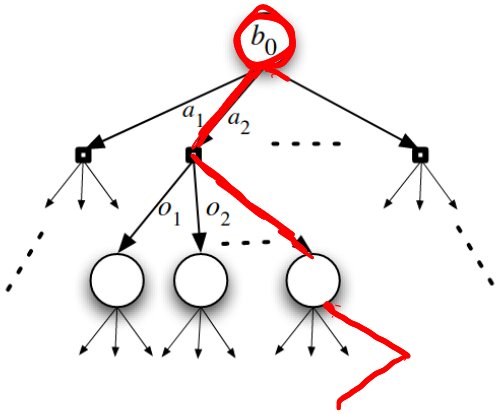
Kurniawati H, Hsu D, Lee WS. SARSOP: Efficient Point-Based POMDP Planning by Approximating Optimally Reachable Belief Spaces.

SARSOP

Successive Approximation of Reachable Space under Optimal Policies

(One instantiation of PBVI)

- Start with $B = \{b_0\}$, Γ = blind lower bound
- Maintain B with a belief tree
- Maintain value bounds (Γ: lower, "sawtooth": upper)
- At each iteration, sample new belief points to add to tree
  - Choose actions with upper bound
  - Choose observations with upper-lower



- Can solve problems with up to ~100,000 states
- Hidden implementation details :(((
- HSVI - similar algorithm, better paper

Kurniawati H, Hsu D, Lee WS. SARSOP: Efficient Point-Based POMDP Planning by Approximating Optimally Reachable Belief Spaces.

SARSOP

Successive Approximation of Reachable Space under Optimal Policies

(One instantiation of PBVI)

- Start with $B = \{b_0\}$, Γ = blind lower bound
- Maintain B with a belief tree
- Maintain value bounds (Γ: lower, "sawtooth": upper)
- At each iteration, sample new belief points to add to tree
  - Choose actions with upper bound
  - Choose observations with upper-lower
- Prune
  - Tree and B to stay in $\mathcal{R}^*(b_0)$
  - α vectors not dominant for any $b \in B$

Terminates when upper and lower bounds at b_0 less than ε

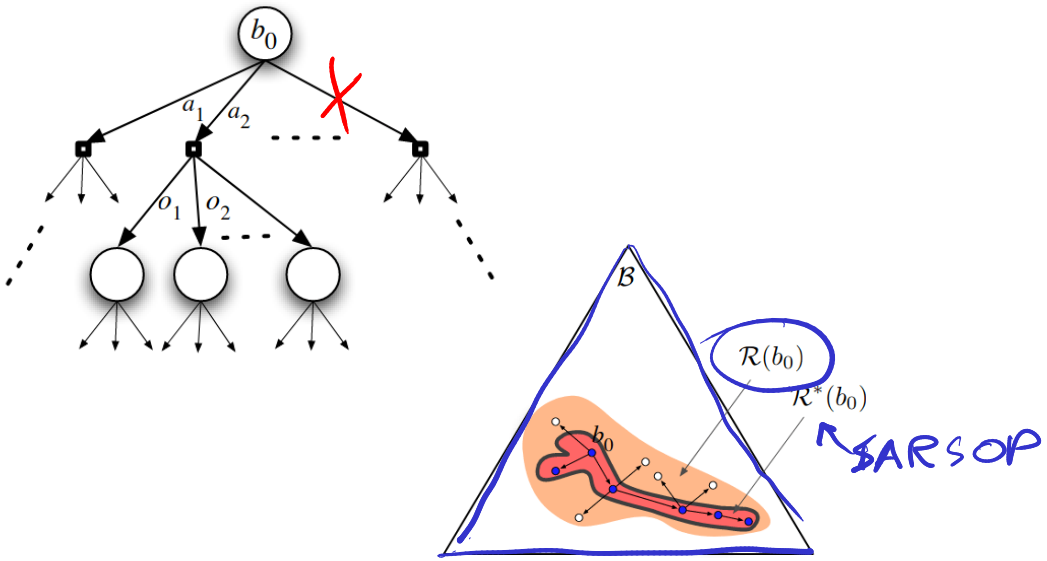


Fig. 1. Belief space $\mathcal{B}$, reachable space $\mathcal{R}(b_0)$, and optimally reachable space $\mathcal{R}^*(b_0)$. Note that $\mathcal{R}^*(b_0) \subseteq \mathcal{R}(b_0) \subseteq \mathcal{B}$.

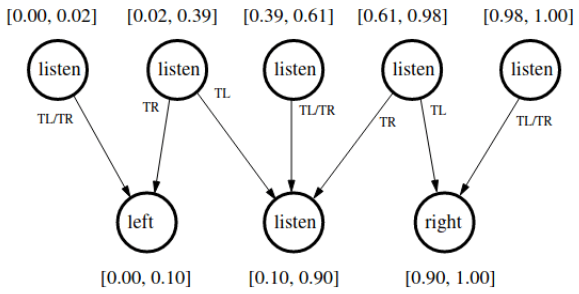
- Can solve problems with up to ~100,000 states
- Hidden implementation details :(((
- HSVI - similar algorithm, better paper

Kurniawati H, Hsu D, Lee WS. SARSOP: Efficient Point-Based POMDP Planning by Approximating Optimally Reachable Belief Spaces.

Tiger POMDP Solution

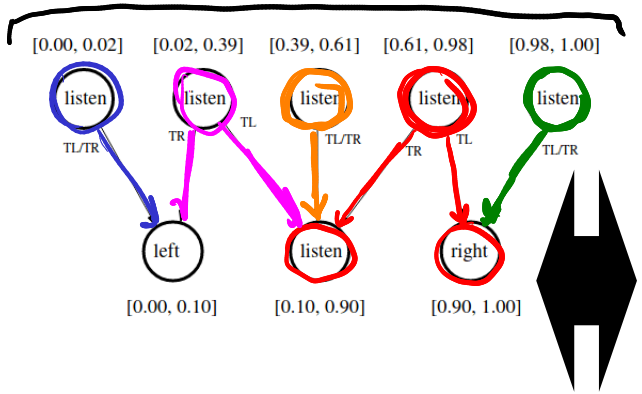
Tiger POMDP Solution

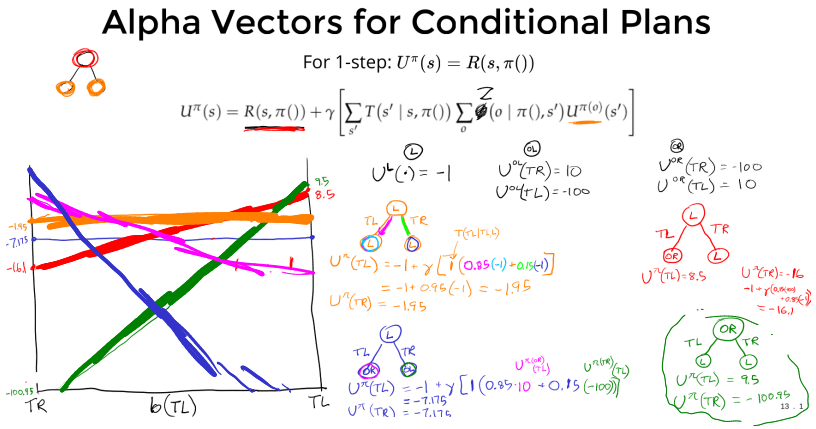
(Undiscounted) from Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence*



Tiger POMDP Solution

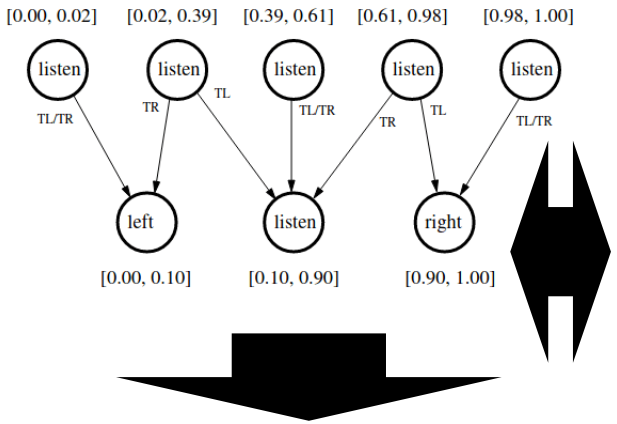
(Undiscounted) from Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence*

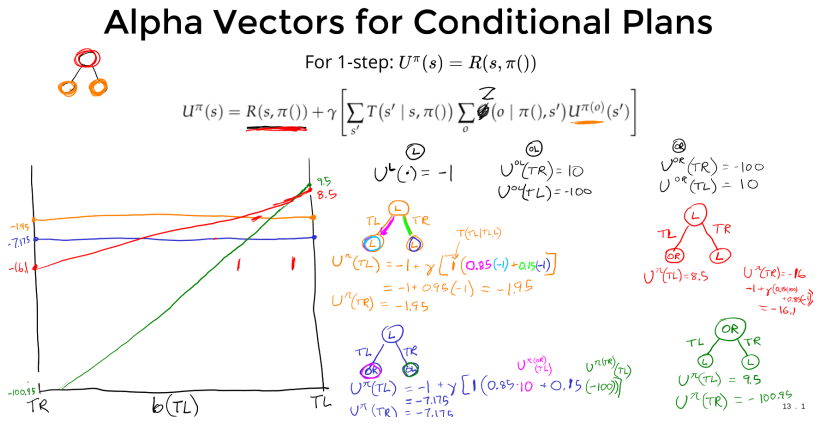


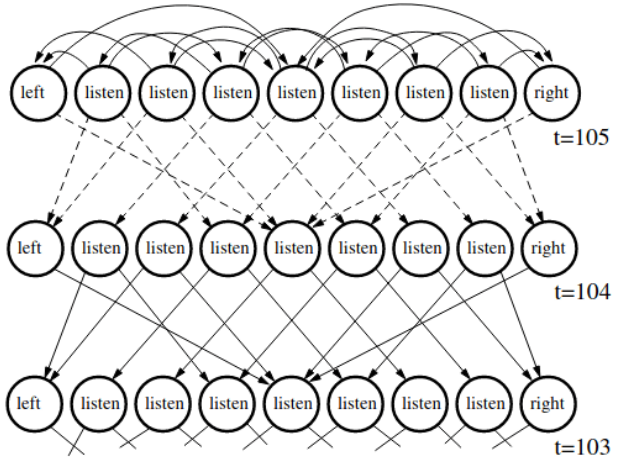


Tiger POMDP Solution

(Undiscounted) from Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence*

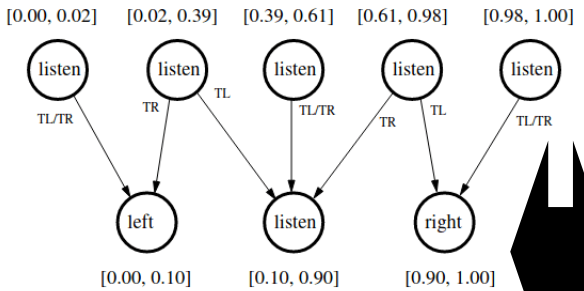


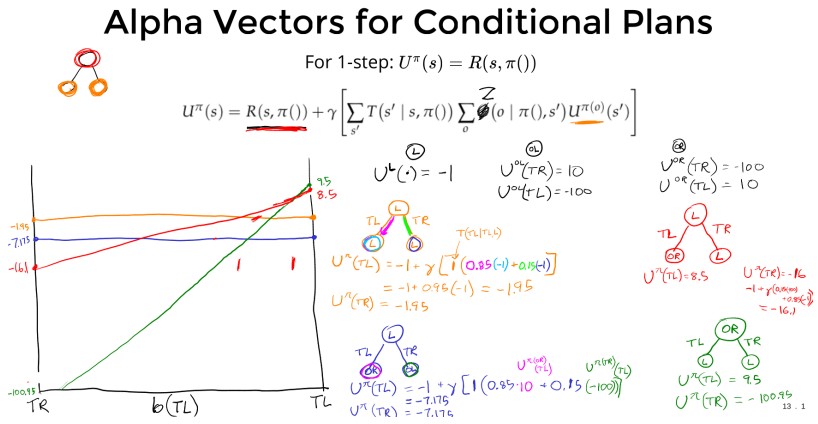


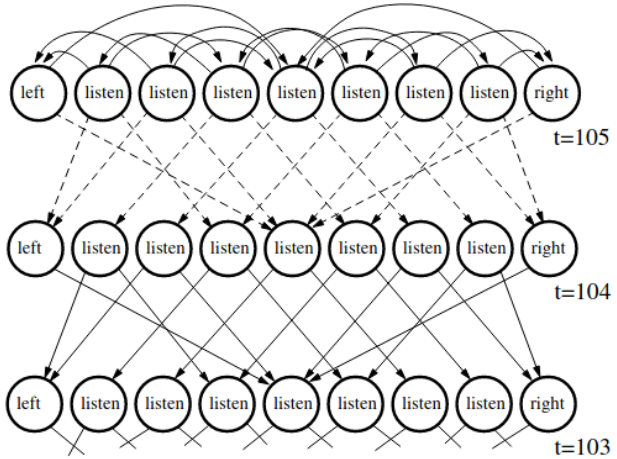


Tiger POMDP Solution

(Undiscounted) from Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence*

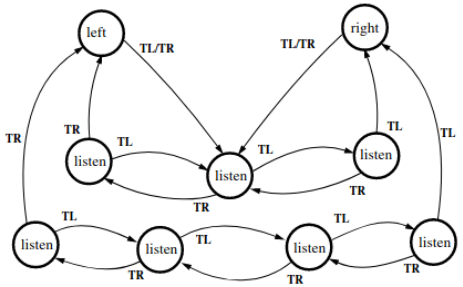






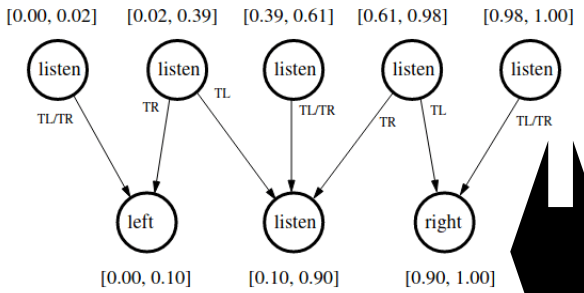
"Policy Graph" /

"Finite State Controller"



Tiger POMDP Solution

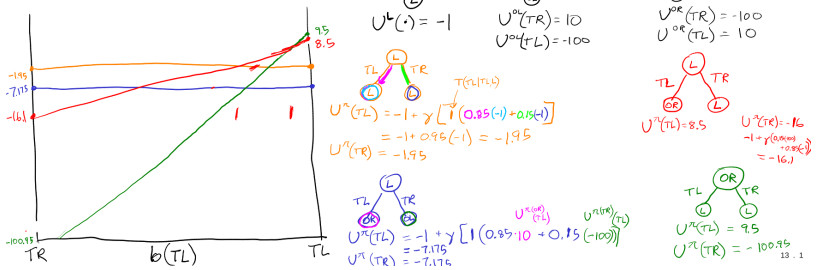
(Undiscounted) from Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence*

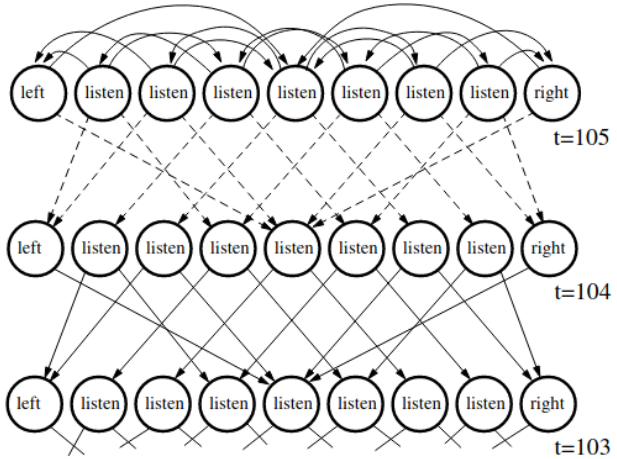


Alpha Vectors for Conditional Plans

For 1-step: $U^\pi(s) = R(s, \pi())$

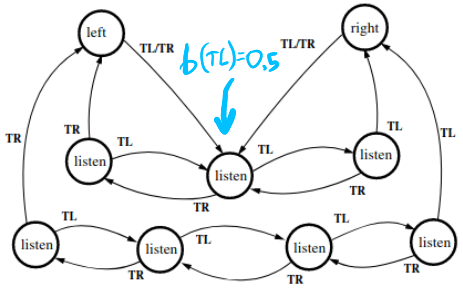
$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_o Z(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

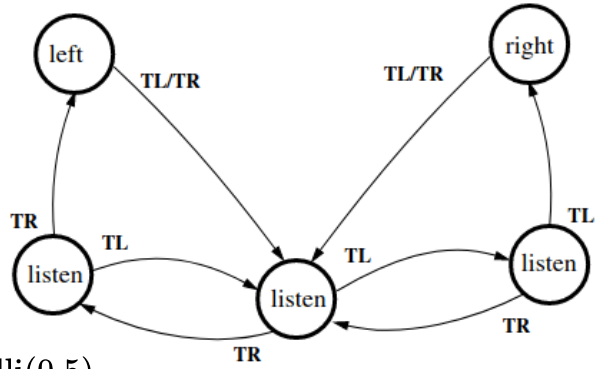




"Policy Graph" /

"Finite State Controller"

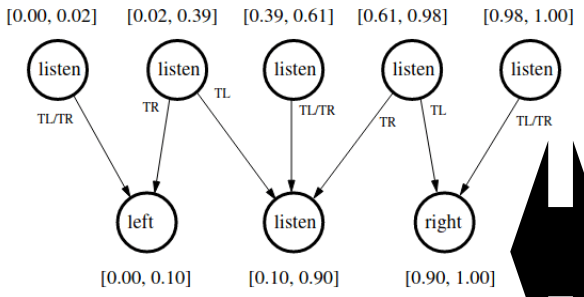


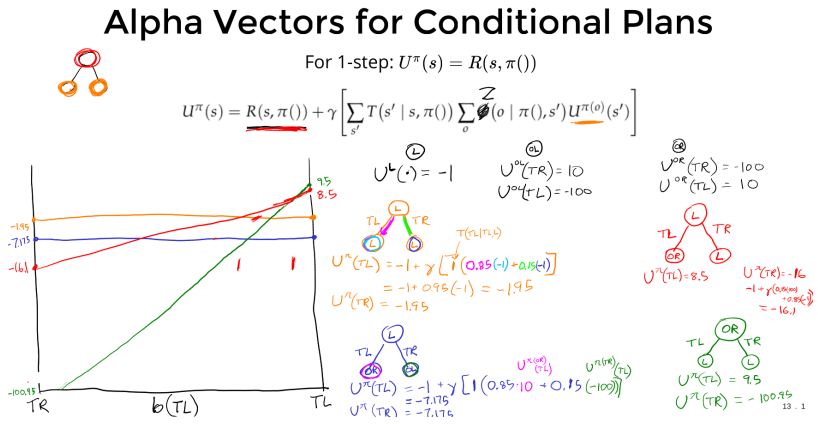


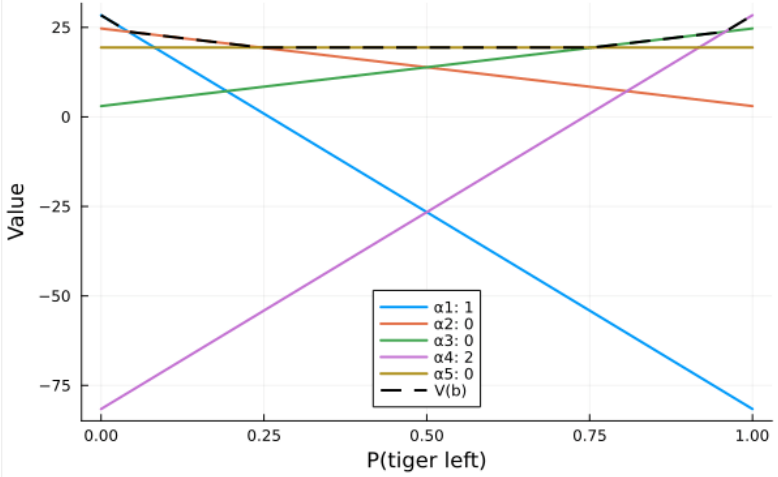
Assuming $b_0 = \text{Bernoulli}(0.5)$

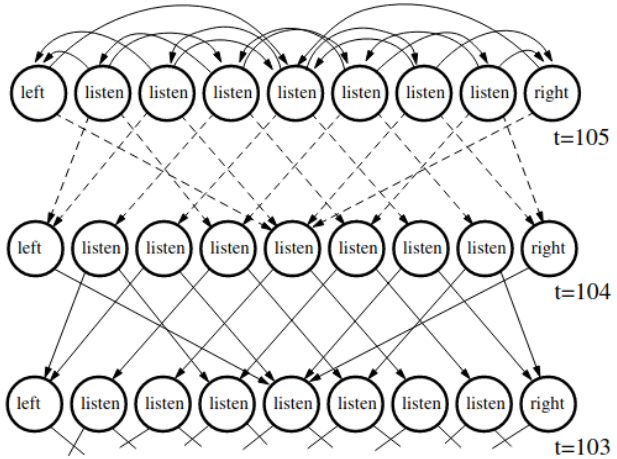
Tiger POMDP Solution

(Undiscounted) from Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence*



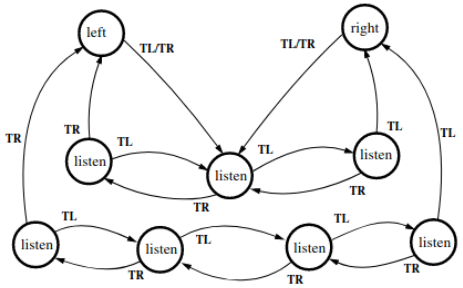


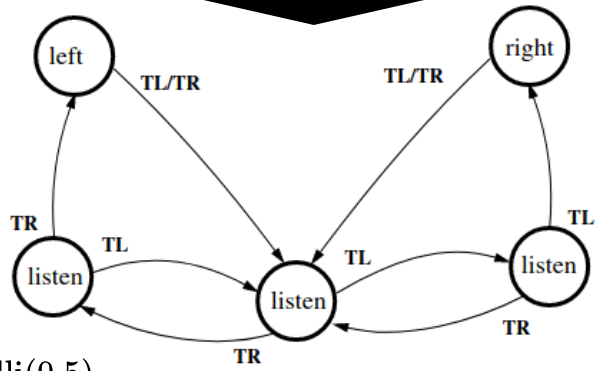




"Policy Graph" /

"Finite State Controller"





Assuming $b_0 = \text{Bernoulli}(0.5)$